

Challenge (Habanero)

- Q1.** A gamer's score is $\frac{1}{\sqrt{2}}$ times their opponent's score. If the opponent's score is 16, what is the gamer's score after rationalizing the denominator?
(A) $4\sqrt{2}$
(B) $8\sqrt{2}$
(C) 4
(D) 8
(E) $2\sqrt{2}$
- Q2.** A CPU uses $\frac{1}{\sqrt{3}}$ times the power of a GPU to process data. If the GPU uses 12 watts, what is the power used by the CPU after rationalizing the denominator?
(A) $2\sqrt{3}$ watts
(B) $4\sqrt{3}$ watts
(C) $6\sqrt{3}$ watts
(D) 2 watts
(E) 4 watts
- Q3.** A gamer's data usage is $\frac{1}{\sqrt{5}}$ times their average data usage. If the average data usage is 20 GB, what is the gamer's data usage after rationalizing the denominator?
(A) $2\sqrt{5}$ GB
(B) $4\sqrt{5}$ GB
(C) 2 GB
(D) 4 GB
(E) $\sqrt{5}$ GB
- Q4.** A coding logic uses $\frac{1}{\sqrt{6}}$ times the number of loops as a similar code. If the similar code has 18 loops, what is the number of loops in the coding logic after rationalizing the denominator?
(A) $3\sqrt{6}$
(B) $3\sqrt{2}$
(C) 3
(D) $\sqrt{6}$
(E) $\sqrt{2}$
- Q5.** A scoreboard displays $\frac{1}{\sqrt{8}}$ times the score of a player. If the player's score is 24, what is the score displayed on the scoreboard after rationalizing the denominator?
(A) $3\sqrt{2}$
(B) $6\sqrt{2}$
(C) 3
(D) 6
(E) $\sqrt{2}$
- Q6.** A tech company uses $\frac{1}{\sqrt{10}}$ times the number of employees as a similar company. If the similar company has 30 employees, what is the number of employees in the tech company after rationalizing the denominator?
(A) $3\sqrt{10}$
(B) $\sqrt{10}$
(C) 3
(D) $\sqrt{10}$
(E) 3
- Q7.** A gamer's reaction time is $\frac{1}{\sqrt{12}}$ times their opponent's reaction time. If the opponent's reaction time is 10 seconds, what is the gamer's reaction time after rationalizing the denominator?
(A) $\frac{5}{\sqrt{3}}$ seconds
(B) $\frac{10}{\sqrt{3}}$ seconds
(C) $\frac{5}{\sqrt{6}}$ seconds
(D) $\frac{10}{\sqrt{6}}$ seconds
(E) $\frac{5}{\sqrt{2}}$ seconds
- Q8.** A CPU processes $\frac{1}{\sqrt{15}}$ times the data of a GPU. If the GPU processes 25 bytes, what is the data processed by the CPU after rationalizing the denominator?
(A) $\frac{5}{\sqrt{15}}$ bytes
(B) $\frac{5}{\sqrt{3}}$ bytes
(C) $\frac{5}{\sqrt{5}}$ bytes
(D) $5\sqrt{3}$ bytes
(E) $5\sqrt{15}$ bytes

Open Ended Questions (FRQ)

- Q9.** A coding logic uses $\frac{2}{\sqrt{3}}$ times the number of loops as a similar code. If the similar code has x loops, what is the number of loops in the coding logic after rationalizing the denominator? Show your work and simplify your answer.
- Q10.** A tech company uses $\frac{3}{\sqrt{5}}$ times the number of employees as a similar company. If the similar company has y employees, what is the number of employees in the tech company after rationalizing the denominator? Show your work and simplify your answer.